TerraAlert v3 landslide model: how it works, what we are improving, and why

A learning guide written as we go. Each step says what we did, why, and what the numbers mean. Keep it with the project so future changes can be checked against the same plan.

1. What the model does

The model looks at one place on one day and gives a risk score from 0 to 1. It uses 20 inputs: rain over the last 3, 7, 14 and 30 days, slope, elevation, aspect, vegetation (NDVI), moisture (NDWI) and soil type, plus a few combinations of these. It was trained on **552 recorded landslides** and, for each one, two kinds of "safe" examples: a nearby slope on the same day, and the same spot in a normal year. The score is turned into a warning level (Low, Moderate, High, Critical) by cut-offs chosen from honest test scores.

2. Words you need

Term	Plain meaning
Accuracy	Share of all places called right (safe or landslide). Misleading here: 2 of 3 places are safe, so never warning scores 66%.
Recall (catch rate)	Share of real landslides that got an alert. 70% means 7 in 10 caught.
Precision	Share of alerts that were followed by a real landslide.
False alarm rate	Share of safe places that got an alert wrongly.
Balanced accuracy	Average of recall and "safe places cleared". Fair when safe places are far more common.
ROC-AUC	How well the score ranks risky places above safe ones, from 0.5 (guessing) to 1.0 (perfect). Cut-off free.
Out-of-fold test	Each area is scored by a model that never saw it while training.
Overfitting	Learning the training rows by heart. Shows as a big gap between training and test scores.
Data leakage	Test information sneaking into training. It makes scores look better than they are.
Time-based test	Train on older years, test on newer years. Checks that the model works on future events.
Cut-off (threshold)	The score above which we raise an alert. Moving it trades missed landslides against false alarms.

3. Where the model stands today

Measure (honest out-of-fold test, 1,610 rows)	Value
Accuracy at the MODERATE alert level	54.7%
Balanced accuracy	58.4%
Landslides caught (recall)	69.9%
Alerts that were real (precision)	40.7%
ROC-AUC	0.613

Reading these: the model catches about 7 in 10 past landslides, but about half of its alerts are false alarms. Accuracy looks low mainly because most places are safe. This is a useful early warning, not a precise forecast.

4. The plan and where each step stands

Step	What it means	Status
1. Better landslide records	Add the Himachal landslide list with usable dates and locations.	CLEANED. 459 events with a real year kept from 6539 rows. Only 8 have a full day, so same-day controls are not possible. Details below.
2. Ground inputs (geology, soil moisture)	Add rock type and soil moisture the model cannot see today.	Needs your approval: it changes the 20-input list the app sends, which you asked us to keep.

Step	What it means	Status
3. Honest time-based test	Train on 2007-2013, test on 2014-2016.	DONE. ROC-AUC 0.600, recall 64.7% on 167 later landslides.
4. Alert level choice	Pick the trade-off between missed landslides and false alarms.	Policy decision with HPSDMA. Options are listed below.

Step 1 in detail: cleaning the Himachal landslide list

What we received: a file of 6,539 Himachal landslide rows (hp_landslides.csv). Every row is a landslide, so there are no safe examples in it. The file is kept unchanged in ai-module/data/raw/.

What we did, in order (script: ai-module/v3/build/clean_hp_landslides.py):

1. Tidied the text. Removed spaces at the ends and turned placeholders ('#N/A', blank, '0') into blanks.

2. Kept rows with a numeric latitude and longitude. Dropped: 0.

3. Kept rows inside a Himachal box (lat 30.3-33.3, lon 75.5-79.0). Dropped: 10.

4. Kept rows with a real year (1900-2026). The year '0' was treated as missing, not as a date. Dropped: 6047. This was the biggest loss.

5. Read a full day from the history text only when it was written out, such as '14th August 2023'. Found in 8 rows.

6. Removed duplicates: same year and same place (coordinates to 3 decimals, about 100 m). Dropped: 23.

Result: 459 landslides saved to ai-module/data/v3/hp_landslides_clean.csv, years 1970-2023. Before 2014: 166; 2014-2016: 152; 2017 onward: 141. The 2015 count is high (93), which may be one reporting batch; this should be checked before use.

What this means for the model: the file has real locations and years, so it can feed the time-based test. It cannot yet build the same-day controls, because almost every row has only a year. It also has no safe examples. So step 1 is cleaned but not yet used to retrain the model. Retraining would need controls built from the cleaned events, and the 20 inputs would stay the same.

Step 3 in detail

We trained only on events before 2014 (385 landslides) and tested on 2014-2016 (167 landslides). The landslide and its controls always go into the same period, so no event is split. The results were: ROC-AUC 0.600 (spatial test 0.613), recall 64.7% (spatial 70%), balanced accuracy 57.3%. The model keeps most of its skill on later years.

Caveats we must state: (a) the test has only about 170 landslides, so each number has a margin of error of a few points; (b) the alert level was chosen using all years, so this test is slightly kinder to the model than a fully fresh test would be.

Step 4 options for the alert level (for HPSDMA to decide)

Option	Effect (from the honest test)	Trade-off
Lower cut-off (more alerts), about 0.43	About 79% caught, about 66% of safe places alerted	Warns more often; people may stop listening
Current cut-off, 0.47 (MODERATE)	About 70% caught, about 53% of safe places alerted	Balanced choice used in the app today
Higher cut-off (fewer alerts), about 0.52	About 50% caught, about 33% of safe places alerted	Quieter, but misses half of landslides

5. Lessons so far (for future work)

- Never judge a model by accuracy alone when one outcome is rare. Use recall, false alarms and balanced accuracy.

- Test on areas and years the model has never seen. A score on rows it trained on is not evidence.

- Choose settings and cut-offs with inner tests, then report one untouched final test.

- Do not change the test to reach a target number. Target 80% accuracy would need a test that hides the hard cases.

- Check each change against the same honest test. If it does not help there, it does not go into the app.

- Data quality limits the ceiling. Better labels and ground inputs matter more than tuning.

6. How to reproduce the numbers

```
py -3.11 ai-module/v3/build/clean_hp_landslides.py          # step 1: clean the Himachal landslide list
py -3.11 ai-module/v3/evaluate/evaluate_v3_like_v2.py       # accuracy table, ROC, confusion matrix
py -3.11 ai-module/v3/evaluate/time_split_test_v3.py        # step 3: time-based test
py -3.11 ai-module/v3/evaluate/research_evaluation_v3.py    # full report (about 5 min)
py -3.11 -m pytest ai-module/v3/evaluate/test_model_evaluation.py -v -s
```

Outputs are written to ai-module/v3/results/. Model files are rebuilt by ai-module/v3/train/.